

MUHAMMAD HUZAIFA AFTAB

Mechanical Engineering Undergraduate | CSWE | CAD/CAE/CFD | Research & Automation

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📁 Portfolio: [muhammadhuzaifaftab.portfolio](https://muhammadhuzaifaftab.github.io) | English (Fluent), German (Elementary)

PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate at UET Taxila and Certified SOLIDWORKS Expert (CSWE), with experience in CAD/CAE/CFD, engineering automation, mathematical modeling, and research. Proficient in SOLIDWORKS, ANSYS, MATLAB, Python, and engineering design. Currently leading an HEC-NRPU-funded research project on EDM machining of aerospace alloys using green dielectric fluids, with additional experience in AI for Additive Manufacturing research, CAD automation, and industrial engineering.

EDUCATION

B.Sc. Mechanical Engineering | University of Engineering & Technology (UET) Taxila, Pakistan

Aug 2023 – July 2027 (Expected) | CGPA: 3.38/4.00 | 104 Credit Hours Completed

Relevant Coursework: Machine Design, CAD/CAE, Manufacturing Processes, Mechanics of Materials, Materials Engineering, Measurement & Instrumentation, Control Engineering, Thermodynamics, Fluid Mechanics, Numerical Analysis, Programming Fundamentals

TECHNICAL SKILLS

- **CAD & Automation:** SOLIDWORKS (Certified Expert – CSWE), SOLIDWORKS API/VBA Macros, SOLIDWORKS PDM Add-ins, Solid Edge (Certified Mechanical Expert), GD&T, Engineering Drawings, Reverse Engineering, Assembly Design
- **Simulation & Analysis:** ANSYS (FEA, Fluent/CFD), Structural & Thermal Analysis, Mesh Generation & Convergence Studies, Nonlinear Analysis, HAP (HVAC Load Simulation)
- **Mathematical & Analytical Modeling:** Fluid Machinery Performance Analysis, Parametric Engineering Calculations, Data Analysis & Visualization
- **Programming & Control:** Python, MATLAB, VBA, C# (SOLIDWORKS API/PDM), Arduino, PID Control Systems
- **Manufacturing & Additive Manufacturing:** Machining, CNC, Foundry, Forging, Additive Manufacturing/3D Printing, Welding, Production Planning, Quality Assurance/QC, Experimental Testing & Validation
- **Research & Documentation:** Literature Review, Technical Writing, Engineering Reports

RESEARCH EXPERIENCE

Final Year Research Project | Group Leader

EDM Machining of Aerospace Alloys Using Green Dielectric Fluids (HEC-NRPU Funded Project), UET Taxila | 2026 – Present

- Lead a four-member undergraduate research team investigating electrical discharge machining (EDM) of aerospace alloys using environmentally friendly green dielectric fluids, under faculty supervision, as part of a Higher Education Commission (HEC) National Research Program for Universities (NRPU)-funded initiative.
- Coordinate experimental planning, literature review, and technical documentation as group leader, applying structured research methodology to an advanced manufacturing and sustainable machining problem.

Research Intern | Group Leader, Manuscript Team

MultiOmics —Remote | 2026 – Present

- Lead a review manuscript team on “AI Applications in Additive Manufacturing: Opportunities, Challenges, and Future Trends,” coordinating research tasks and producing production-quality technical documentation ahead of journal submission.
- Conduct systematic literature review, critical analysis of peer-reviewed sources, citation verification, and technical writing to support evidence-based engineering argumentation.

PROFESSIONAL EXPERIENCE

Engineering Operations Intern

Wah Nobel | Aug 2026 – Present | *Multinational joint venture between Swedish and Saudi Arabian partners*

- Gaining hands-on exposure to chemical plant operations, manufacturing processes, industrial equipment, and plant utilities, including pumps, valves, and piping systems.
- Developing practical understanding of mechanical systems, industrial maintenance practices, process equipment, and safety procedures within a chemical manufacturing environment; collaborating with technical staff, observing engineering operations, and preparing technical documentation of plant activities.

Product Design Intern

Machadav | Jul 2026 – Aug 2026

- Completed a 4-week hands-on internship focused on product development, PCB enclosure design, CAD modeling, and 3D-printing prototyping; designed and refined PCB enclosures and mechanical components in CAD, optimizing for functionality, manufacturability, and 3D printing.
- Evaluated and developed mechanical mechanism concepts for product improvement, translating requirements into practical CAD designs.

Manufacturing Operations Intern

Heavy Mechanical Complex (HMC), Taxila | Jul 2025 – Aug 2025

- Completed a six-week rotational internship at HMC, Pakistan's "Mother Engineering Industry," gaining hands-on exposure to production planning, quality inspection, foundry, forging, CNC machining, and assembly.
- Developed practical understanding of large-scale manufacturing, precision engineering, and industrial production workflows, connecting mechanical design concepts with real-world manufacturing processes.

CAD/CAE Design Intern

National Aerospace Science & Technology Park (NASTP) | Jun 2025 – July 2025

- Modeled aerospace components and assemblies in SOLIDWORKS, maintaining tight dimensional accuracy for downstream manufacturability.
- Performed structural/FEA simulations in ANSYS to validate design integrity under load, supporting design optimization decisions.

Freelance CAD Design Engineer

Fiverr / Upwork — Remote | Aug 2024 – Present

- Delivered 25+ end-to-end CAD and mechanical design projects for international clients, managing technical scoping, design development, and production-ready documentation from initial requirements through final delivery.
- Completed precision 2D-to-3D CAD modeling and reverse engineering for client assemblies, including a screen lens/sealing assembly, ensuring design intent translated accurately into manufacturable geometry.

SELECTED ENGINEERING PROJECTS

Programming & Automation | SOLIDWORKS API/VBA, Python, MATLAB

Personal & Academic Projects

- Developed parametric SOLIDWORKS macros using the SOLIDWORKS API (VBA) to automatically generate small mechanical components (e.g., shafts) and their production drawings; completed the CAD Sharp LLC SOLIDWORKS VBA & API Programming Course covering macro development, the API object model, and automated part/assembly/drawing creation.
- Applied SOLIDWORKS PDM API concepts for add-in development (certified in PDM API Write Add-ins in C# and VBA for Non-Programmers); built a personal engineering portfolio website, contributed to a fee-verification web app in Python, and used MATLAB for engineering data analysis and visualization.

Automated Water Level Control System (PID Controller)

Complex Engineering Problem, UET Taxila | Arduino, MATLAB/Simulink, PID Control

- Developed an Arduino-based PID water-level controller with ultrasonic sensing and automated pump control, targeting 8 cm with a 3–8 cm operating range.
- Tuned $K_p = 2$, $K_i = 0.05$, $K_d = 0.01$ and achieved zero overshoot with ~236 s settling time in MATLAB/Simulink.

Centrifugal Pump — Effect of Impeller Inlet Vane Angle on Performance

Fluid Machinery | Analytical & Mathematical Modeling

- Investigated the effect of impeller inlet vane angle (16° – 30°) on centrifugal pump performance using a simplified analytical model, calculating outlet whirl velocity, impeller power, and pump efficiencies at each angle.
- Found that increasing the angle reduced outlet whirl velocity ($18.90 \rightarrow 12.59$ m/s) and impeller power ($95 \rightarrow 63.2$ kW), while increasing manometric efficiency ($61.8\% \rightarrow 93.0\%$).

Electrically Actuated Spring-Loaded Centrifugal Governor

Machine Design-II | SOLIDWORKS & COMSOL Multiphysics

- Designed and analyzed a 500–3000 RPM centrifugal governor using SOLIDWORKS, combining mechanical speed regulation with electrical actuation and a spring-damper system.
- Performed theoretical and COMSOL analysis, obtaining 5,919 N centrifugal force per flyball, 0.033–0.038 m sleeve displacement, and 75.4 MPa arm stress, confirming stable motion and safe operation

Battery Pack Heat Transfer Analysis

ANSYS Fluent | Heat & Mass Transfer

- Developed a CFD model of 4 prismatic lithium-ion cells with water cooling, using 250 W/m^3 heat generation and 0.05 m/s inlet velocity.
- Achieved 6–8% lower maximum temperature ($\sim 53 \rightarrow 49$ – 50°C) by increasing cell spacing, with <2% mesh sensitivity and 10–15% correlation deviation.

Cam-Driven Color Sorting Mechanism

Mechanism Design & Kinematic Analysis, UET Taxila

- Designed and tested an Arduino-based cam-follower system for automatic red/green color sorting, using SHM cam profiles and 2 independent 1-DOF mechanisms.
- Achieved 40 mm follower stroke, 125.7 mm/s maximum velocity, 789.6 mm/s^2 acceleration, and 95% sorting accuracy, with a safe 18.4° pressure angle.

LEADERSHIP & COMMUNITY

Founder & Lead Mentor

MechMate Hub | Aug 2024 – Present

- Founded and grew a global engineering learning community of 600+ mechanical engineering students, providing structured SOLIDWORKS certification roadmaps, resources, and live mentoring.
- Directly guided 50+ students and engineers through SOLIDWORKS certification preparation, with successful exam outcomes.

Chairperson & Lead Mentor

SOLIDWORKS Club, UET Taxila | Jul 2024 – Present

- Founded and lead a 40+ member volunteer club organized into five specialized teams, bridging academic CAD theory with real-world industry practice for engineering students
- Organize and host technical workshops and webinars connecting industry professionals with students, while personally mentoring members in SOLIDWORKS and practical CAD/design workflows

Campus Ambassador

ASPIRE Institute & Siemens | Aug 2026 – Present

- Represented ASPIRE Institute as a Campus Leader, engaging in global leadership and fellowship programs alongside university students worldwide
- Served as a Siemens AG Campus Ambassador, promoting engineering software, technology, and career opportunities within the student community.

Assistant Vice Chair

ASME UET Taxila Student Chapter | Nov 2023 – May 2026

- Coordinated chapter events and activities, served as a speaker at technical and professional sessions, and supported collaboration within the engineering community.

Media Secretary

UMEED-E-SUBH Welfare Society, UET Taxila | Aug 2023 – Present

- Progressed through leadership roles in Media & Coverage and CSP, supporting fundraising, project coordination, and volunteer activities.
- Managed media coverage and social media for welfare initiatives, including Ramadan, Winter Uniform, and Stationery Drives.
- Contributed to education and community projects supporting underprivileged students and women.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Certified SOLIDWORKS Expert (CSWE) with 20+ SOLIDWORKS Certifications — Dassault Systèmes
- Certified DriveWorksXpress Associate (CDWXA)
- Solid Edge Mechanical Expert Certification — Siemens
- Completed MATLAB Onramp and Simulink Onramp — MathWorks
- Geometric Dimensioning & Tolerancing (GD&T) for Engineering — Udemy
- Google Project Management Professional Certificate V2 — Google / Coursera
- Certified Istec Vibration Training Category 1+ — ISTE International

Conferences Attended

- 3rd Global Science, Technology and Management Conference (GSTMC-25), University of Jhang — Jan 2025
- 1st International Conference on Advances in Mechanical, Material, Mechatronics and Energy Engineering (ICAME-24)
- 2nd UW-POFs Joint Conference on Academia's Collaboration with Industrial Organizations (CACIO-2022) — Dec 2022

AWARDS & RECOGNITION

- Youngest Certified SOLIDWORKS Expert (CSWE) and SOLIDWORKS Champion in Pakistan — Dassault Systèmes
- Top 10 Performer, ASPIRE Pakistan Founders Edge Cohort III — National Entrepreneurship Recognition
- Best Assistant Vice Chair of the Month — ASME UET Taxila Student Chapter

PROFESSIONAL MEMBERSHIPS

- SOLIDWORKS Champion, Dassault Systemes Global Network
- Affiliate Member, Institution of Mechanical Engineers (IMechE)
- Student Member, American Society of Mechanical Engineers (ASME)
- Member, International Association of Engineers (IAENG)

ADDITIONAL TRAINING

- McKinsey Forward Program — 10-Week Professional Skills Program, McKinsey & Company
- Low Emission Analysis Platform (LEAP) — Intensive 2-Month Training, SDPI
- Entrepreneurship Training — 3.5-Month Program, ASPIRE Pakistan